

KHUSHI KUMARI

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Education

Indian Institute of Technology Kharagpur

Bachelor Of Technology in Civil Engineering; GPA - 9.05/10

Aug 2023 – May 2027

Kharagpur, India

Internship Experience

Strategy Intern | Tata Motors

Jun 2023

Assessing 10+ emerging truck technologies by mapping their maturity, scalability, relevance to Indian commercial markets

- Benchmarked global OEMs and Indian startups to derive insights on innovation, localization, and regulatory trends
- Contributed to Tata Motors' FY30 Technology Roadmap by analyzing policies, fleet expectations, and adoption enablers
- Supported R&D strategy by identifying high-impact focus areas and mitigation strategies through market research

Projects

Traffic Congestion Classifier | Course Project | Prof. Bhargab Maitra

Apr 2025

Built a ML classifier to predict congestion Level using **Bangalore traffic dataset (10+ features)**

- Conducted correlation analysis to select top predictive features and engineered a **multi-class target variable**
- Developed end-to-end pipeline(**imputation, scaling, encoding**) with Random Forest, achieving **91% accuracy**
- Developed an interactive **Streamlit app** for real-time congestion prediction in specific areas using the trained pipeline

News Category Classifier | NLP Classification Project | Self Project

May 2025

Built a news classification system on the **BBC dataset** containing 2,200+ articles across 5 categories

- Processed text with NLTK (**tokenization, lemmatization, stopword removal**) and **TF-IDF** vectorization
- Trained **Logistic Regression** with custom preprocessing and **label encoding**, achieving **96% accuracy**
- Developed an **interactive UI** enabling **real-time** news category predictions from **user-input text**

Fashion MNIST Classifier | CNN-Based Image Classifier | Self Project

Jun 2025

Built a **CNN** with **TensorFlow** on **Fashion MNIST (70,000+ images)**, achieving **88% accuracy**

- Preprocessed grayscale images with normalization and reshaping for consistent model input for training and evaluation
- Tracked training and validation trends via accuracy-loss curves, guiding model tuning and improving transparency
- Analyzed class-wise **precision, recall, F1-scores** to evaluate model strengths & uncover misclassification trends

Building LLMs From Scratch | Vizuarra

Jun 2025

Constructed a **Transformer-based mini-Large Language Model** to understand core NLP building blocks

- Tokenized corpus with **tiktoken** (BPE) and trained **50K+** tokens with attention and feedforward layers
- Built and trained a **12 million+** parameter model on **Harry Potter text** for next-token prediction task
- Evaluated training loss convergence, perplexity & scalability constraints to understand limitations of small-scale LLMs

Relevant Coursework

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|---------------------------------|-------------------------------|-------------------------------|
| • Programming & Data Structures | • Database Systems | • Advanced Calculus |
| • Probability and Statistics | • Machine Learning | • Linear Algebra |
| • Numerical Analysis | • Natural Language Processing | • Data Structures & Algorithm |

Technical Skills

- Languages & Libraries:** Python, C, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch
- Software/Tools:** Jupyter Notebook, Visual Studio Code, MS Excel, MS PowerPoint, Canva, Power BI, GitHub

Extracurricular

Inter Hall Data Analytics 2024-2025 | IIT Kharagpur

Analyzed multi-source datasets (sales, media spend, NPS, weather) to identify performance drivers Built Power BI dashboards

National Cadet Corps(NCC) | Army Wing Cadet

Served as an NCC Army Wing Cadet for 2 years; earned '**B**' **Certificate** and recognized as **Best Cadet** at CATC

Leadership

Megalith | Civil Engineering Tech Fest

Jul 2024 – Present

Design Team Head

IIT Kharagpur

- Created promotional designs and social media content, reaching **5,000+** students and professionals for Megalith '25
- Managed on-ground coordination for **15+** events, ensuring smooth flow and attendee satisfaction
- Collaborated with **5+** cross-functional teams to plan and implement outreach and promotional strategies